📊🔍 KEY INSIGHTS FROM YOUR PROJECT

🤝 1. CUSTOMER BEHAVIOR INSIGHTS

👉 High-value vs Low-value users

- Some users have **very high avg_spend**, showing premium buying behavior
- Majority of users fall under **Low–Medium spending range**

📌 Insight:

The dataset shows a skewed distribution where a small number of users contribute disproportionately to total revenue.

👉 Purchase Frequency

- Certain users have **high purchase_freq**
- Others are **one-time buyers**

📌 Insight:

Repeat customers are crucial as they generate consistent revenue compared to one-time buyers.

⏳ 2. RECENCY ANALYSIS

- Users with **low days_since_last** → **active users**
- Users with **high days_since_last** → **inactive / churn risk**

📌 Insight:

Recency is a strong indicator of customer engagement. Users inactive for long periods are likely to churn.

💰 3. SPENDING PATTERN (BINNING)

Based on:

- Low (<50)
- Medium (50–150)
- High (>150)

👉 Observations

- Most transactions fall under **Low and Medium**
- Few transactions are **High-value**

📌 Insight:

Customers prefer smaller, frequent purchases rather than large transactions.

📊 4. PAYMENT METHOD ANALYSIS

From your dataset (example):

- Popular methods:
  - UPI
  - Wallet
  - Debit Card

📌 Insight:

Digital payment methods dominate, indicating a shift toward cashless transactions.

📅 5. TIME-BASED INSIGHTS

From extracted features:

- Month / Year patterns visible

👉 **Observations**

- Sales fluctuate over time
- Some months show higher activity

📌 Insight:

Seasonal trends influence purchasing behavior, which can be used for targeted marketing.

📦 6. CATEGORY SPENDING

From cat_spend:

- Users spend differently across categories (Low/Medium/High)

📌 Insight:

Spending behavior varies significantly among users, enabling segmentation into budget vs premium customers.

📉 7. OUTLIER ANALYSIS

- Extreme values removed using IQR

📌 Insight:

Outlier removal improved data consistency and prevented skewed model behavior.

🔄 8. DATA TRANSFORMATION IMPACT

- Log transformation normalized distribution
- Scaling improved comparability

📌 Insight:

Data transformation techniques enhanced model readiness and reduced bias in features.

🤘 9. BUSINESS INSIGHTS (VERY IMPORTANT FOR VIVA)

👉 You can say this confidently:

- Target **high-value users** with premium offers
- Re-engage **inactive users** using discounts
- Focus on **digital payment optimization**
- Promote **medium-range products** (highest demand)

▯ FINAL SUMMARY

The analysis reveals that customer purchasing behaviour is highly varied, with a majority of transactions being low to medium in value. A small segment of users contributes significantly to revenue through high spending and frequent purchases. Recency and frequency are key indicators of customer engagement, while digital payment methods dominate transaction preferences. The dataset shows potential for customer segmentation, targeted marketing, and improved business decision-making.